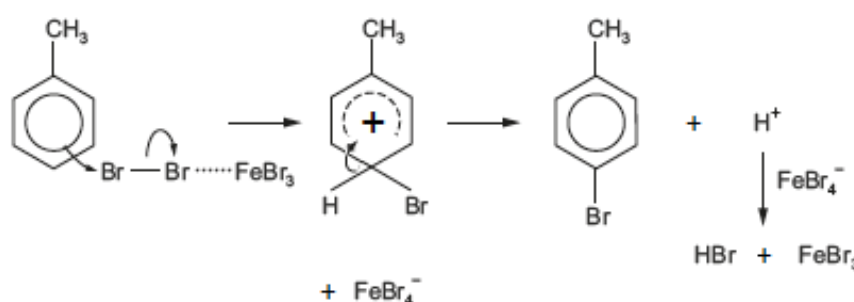


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
13.	(a)	(i)		it is compound N (1) singlet for the —CH₃ group bonded to the carbon atom of the double bond that is also bonded to the bromine atom (2.2δ) (1) the CH proton is as a quartet – split by the adjacent CH₃ protons (5.7δ) (1) the other —CH₃ group is a doublet, split by the adjacent CH proton (1.7δ) (1)			4	4		
		(ii)	I	add aqueous bromine / aqueous acidified manganate(VII) – this is decolourised by L, M and N but unaffected by bromocyclobutane		1		1		1
			II	bromocyclobutane would only give three ¹³C signals (1) L, M and N would each give four signals as each carbon atom is in a different environment for these alkenes (1) accept answer based on C=C at δ 90 to 150 present in L, M and N but not in bromocyclobutane		2		2		
	(b)			 curly arrows (1) partial / full charges (1) regeneration of catalyst (1)	3			3		